

UIU Student Activity tracker

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1 Introduction

This document is a software requirement specification for the UIU Student Activity Tracker. After giving information about the project at the beginning of the document, we will give a complete description for overview and list the requirements which meet the needs of the company.

1.1 Overview

We aim to build a system for the students of UIU, where their official and unofficial achievements' will be tracked and verified. If the student attends any event, seminar or project show then it will be added in his/her profile. This type of profile will guide any faculty in writing a recommendation letter for a student.

2 Motivation

This system will be helpful for the students as well as the alumni of UIU to gather all of their achievements. Students will find it easier in making CVs and resumes. Students will have the incentive to work harder as they will be able to showcase their achievements. To write recommendation letters will be easier for the faculty members.

3 Requirement Study and Analysis

3.1 Present Scenario

Students should be able to keep track of their activities and if there is a chance to get their activities verified it would encourage the student to keep up the good work. United

International University has taken the initiative to keep track of their activities such as projects from different laboratories and internships in different companies. So, they offered the project called Student Activity Tracker for the System Analysis And Design Laboratory to give the students a chance to be a part of it.

3.2 System Study

We gathered information from benchmark products, online resources and questionnaires to conduct the system study. We mainly focused on external sources of information as ours is a new project without an existing user base. From these sources we have fixed our feature list which are described in the Functional Requirements section. Below is a more detailed look at the system study that we have done.

3.3 Benchmark Study

For the benchmark study we have collected data from 4 similar platforms and one online article that shares very similar functionality that we are going to implement. The platforms are:

1. UCAM
2. LinkedIn
3. Github
4. Facebook

Following is our benchmark analysis:

Features	UCAM	LinkedIn	Github	Facebook	Online Article 1	Our Project
1. Profile	✓	✓	✓	✓	✓	✓
2. Update Information	✗	✓	✓	✓	✓	✗
3. Achievements	✗	✓	✓	✓	✓	✓
4. Chat	✗	✓	✗	✓	✓	✗
5. Events	✗	✓	✗	✓	✗	✓
6. Notification	✗	✓	✓	✓	✓	✓
7. Tag	✗	✓	✓	✓	✗	✓
8. Search	✗	✓	✓	✓	✓	✓
9. Data Control	✗	✓	✓	✓	✓	✓
10. Verification	✓	✓	✓	✓	✗	✓

Figure 1: Benchmark Analysis

3.4 Online Article

We studied an online article for a general idea of how a social website should look like. A quick login is a must for every social website. We also found that user interaction can be enhanced using notifications, such as when tagging or sending requests.

3.5 In-Person Interviews and Result Summary

The following is the interview questions and responses we have gathered by interviewing different departments' faculty members.

Interviewer Questions or Objectives	Interviewee Responses
Could you please share your experience when you completed your graduation and made a CV for searching jobs? a. What were the problems you faced? b. What were your expectations? c. Were you satisfied with your process? d. Do you want to change/develop the current system in place?	• No portfolio required • Confusion on making different types of CV • Forgot the achievements from a long time ago • No verification • System should be changed
How do you feel about having a student's achievements in UIU to be displayed on a website?	• It will be a great opportunity for students and filtering them via CV • Motivation and idea for a student
What do you do when you want to attain records/achievements of a sports team or maybe records about a university, etc.? Follow-Up Which websites/apps do you visit most often to attain these records?	• Searching in Google or the different sites according to the topics. • Do manual searching like, asking moderators or admins

Figure 2

What website or data do you use when writing a recommendation letter?	• Courses done under the supervision of the faculty member • Looked into the student's sincerity • Result history from UCAM • Extracurricular Activities like clubs & forums participation (Project Show, MUNs, Debates, etc)
If you had all the achievements of a student stored in one place, would you use that information in any way? Follow-up How would you use that information?	• UA and RA recruitment • Filtering out students for different purposes • Data analysis purpose • Easy to make a recommendation • Club members findings with good experiences • Job history • Easy to collect CV • Moderators for clubs & forums • Interned Students

Figure 3

Would you want achievements posted by students to be verified by someone trustworthy? (Verifying means manually checking individual achievements and giving them a tick) Follow-Up Who do you think should verify students' achievements?	• Yes, that's a good way. • The designated faculty member can verify students' achievements & term-paper • Project shows and other events can be verified by the host or DSA or CCC depending on the events. • Need certificate to verify • Only verified sources of extracurricular activities
How would you feel if some student of yours asked you to verify a certain achievement?	• Would do it willingly, if possible • E-mail notification would be better.
There will be a search bar on our website. What kind of information would you like to search about on the website?	• Filter out based on the requirements • Sub-categories can be helpful • History of events & project shows • Skills • Academic information • Companies under which internships were done

Figure 4

Could you name some types of achievements that you would like to be present on our website?	• Internships, projects, term-paper, MUNs, English participation, soft skills, school achievements, sports • Hardware/ Software tools (circuit simulation, PCB design software, MATLAB, python, microcontroller software, statistical tools, etc)
Is this project a great opportunity for students to gather their achievements in one frame?	• Yes.
What do you think? How can we encourage students to use this site?	• The benefitted outcomes for storing data in one place • Do some kinds of registration in that website • Get recruitment notices • Can apply for UA or RA • Display rankings • Easier & much reflective build up a profile

Figure 5

4 Feasibility Analysis

4.1 Methodology

We chose a questionnaire to gather information to examine a large number of respondents at the same time. We used Google Forms to create our questionnaire.

The following are the demographics of our respondents:

A total of 108 people have responded to our survey until now. Our respondents were male and female students, alumni and teachers of UIU. Most of them were undergraduate students, however, we had some respondents with Honors, Masters and PhD degrees.

Are you connected to any other platforms where you can show your achievements?

101 responses

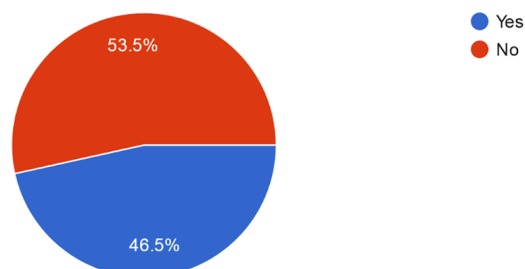


Figure 6

What types of activities do you want us to keep track of? (Select Multiple Answers)

101 responses

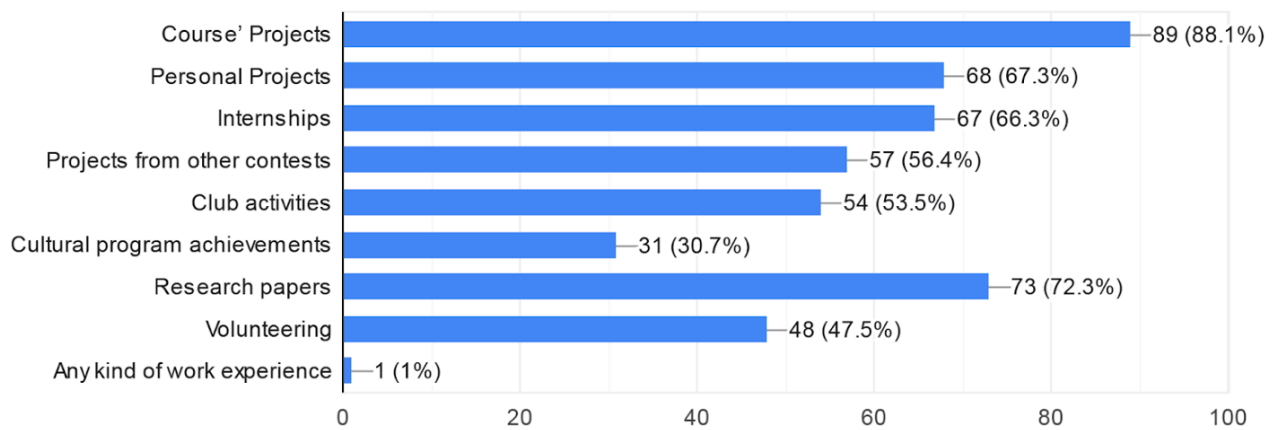


Figure 7

When an event is created, which students on the website should be notified?

100 responses

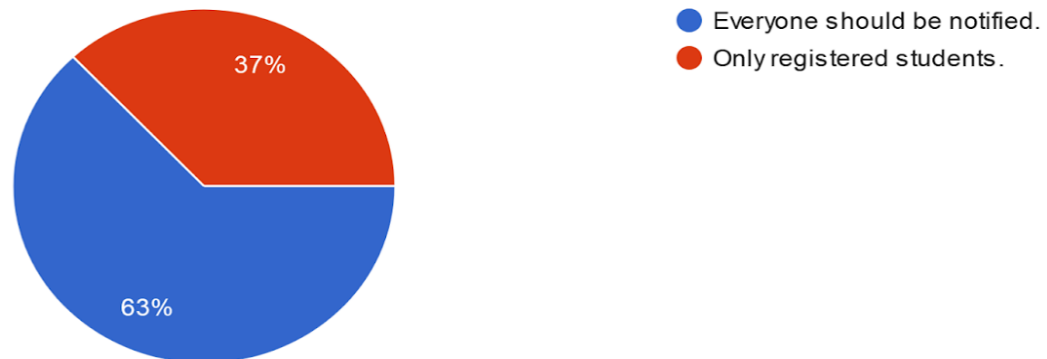


Figure 8

How do you feel about our tag feature (Moderators can tag you in an event that you participated in)?

103 responses

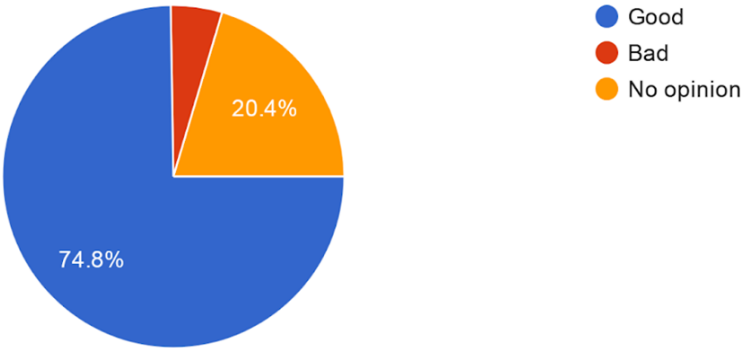


Figure 9

When you upload an achievement, would you want to add keywords with which your achievement can be identified?

101 responses

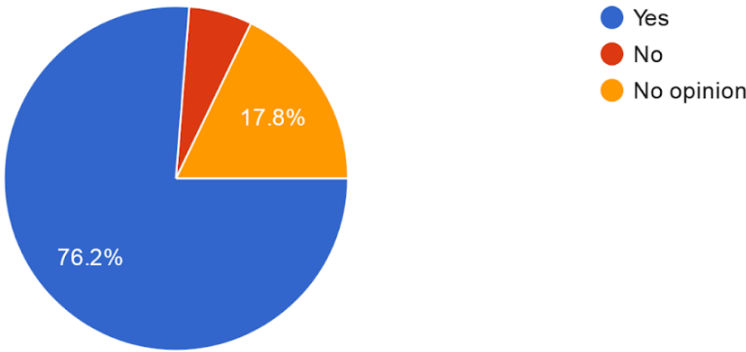


Figure 10

Which website gives you most benefits regarding achievements you post?

93 responses

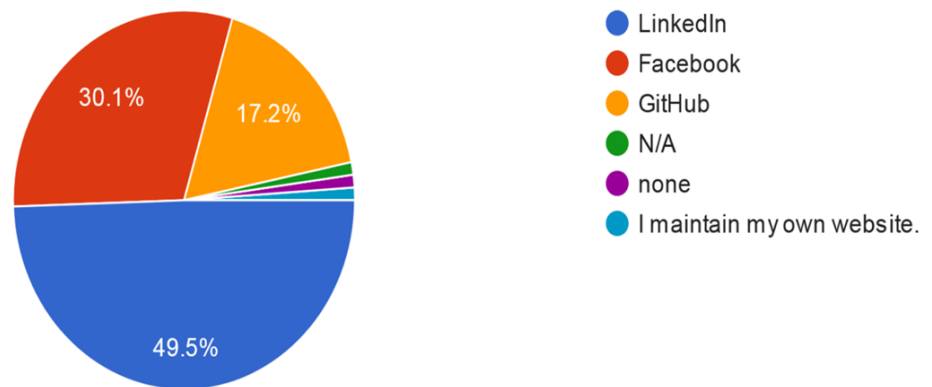


Figure 11

Which website do you mostly use to help you make a CV?

86 responses

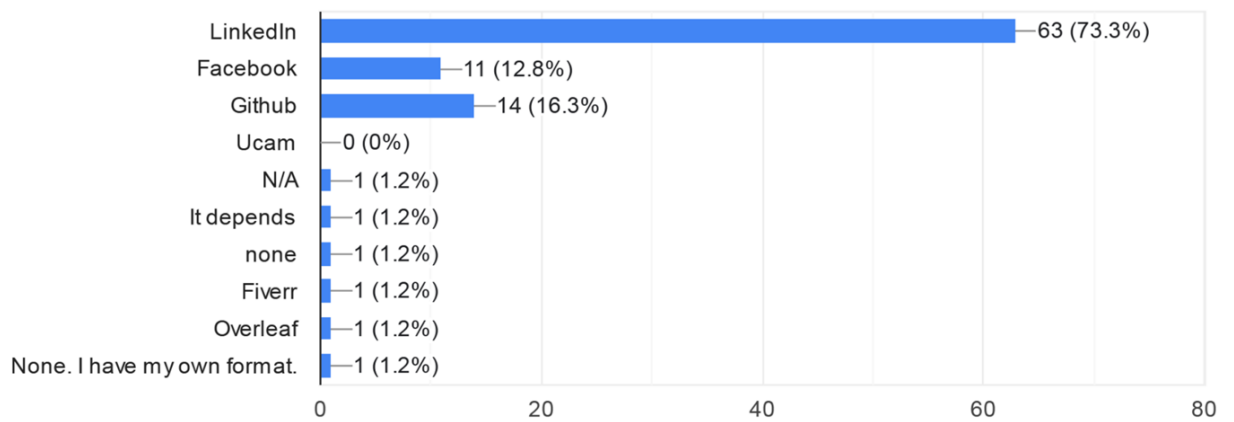


Figure 12

How do you like your profile? 101 responses

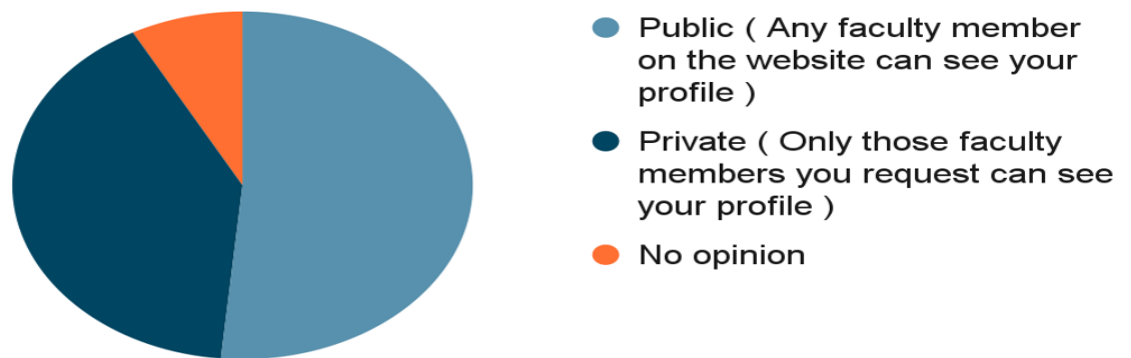


Figure 13

Do you feel like our activity tracker will help you when it comes to recommendation letters?

101 responses

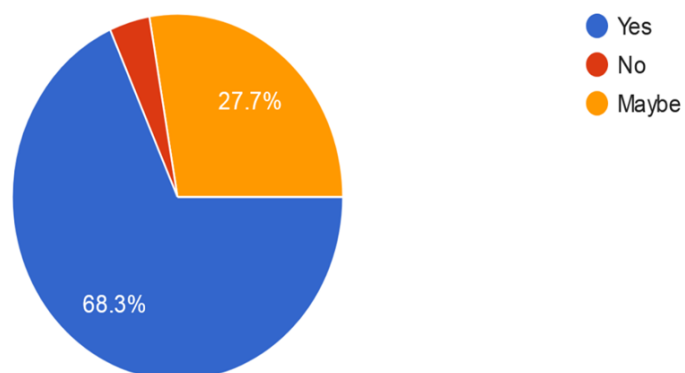


Figure 14

4.2 Survey Results Summary

The amount Of responses in favor of each features from the survey,

1. Profile - 73 percent
2. Show Achievements - 73 percent
3. Events - 63 percent

4. Tag - 75 percent
5. Notification - 64 percent
6. Search - 76 percent
7. Achievement Verification - 85 percent
8. Data Control - 72 percent
9. Generate CV - 80 percent

Extra Feature - 41 percent want their profile to be private (visible to only designated faculty members) so we might think about it.

All of the above proposed features have received positive feedback, so we are going to implement them all.

5 Project requirements

5.1 Analysis results

From all of our survey and Interviews, we have gathered enough information to decide that all of our features are supported by all the users. And we find out the Gap Analysis,

1. Most competitors allow Feature 2 (Update Information), but our system doesn't. This is because platforms such as Facebook and Github are informal and for personal use only, so users can update their information as much as they want. Those information are usually not verified. But our website contains only verified information which are unlikely to be changed anyway can't simply be allowed to be changed by the students.
2. UCAM doesn't have Feature 3 (Achievements), but others as well as our project does have this kind of feature to show their achievements to any faculty member (strength).
3. Most competitors have Feature 4 (Chat), but our system doesn't. We don't need to implement this feature because we found little to no relevant use of messaging in our system.
4. Ucam and GitHub don't have the Feature 5 (Events), but our website will have this feature. In GitHub and Ucam, there is no need for them to create events since they are global, but in our website, we will have different events related to UIU.
5. Ucam doesn't have Feature 6 (Notification) because there is no need for any communication in it. But in our website, there has to be a link between the students and faculty members. There will be various work that will need multiple users' assistance. Because of that we will need a notification feature.

6. UCAM keeps track only of the courses of the students. So, it doesn't have Feature 7 (Tag), but our project keeps track of all achievements. So, we need an event feature.

5.2 Selected Feature Set

Our system will mainly consist of the 9 following features:

1. Profile
2. Show Achievements
3. Events
4. Tag
5. Notification
6. Search
7. Achievement Verification
8. Data Control
9. Generate CV

5.3 Specific Requirements/workflow of the features

Here is a brief description of the functional requirements of the system.

1. Profile:

Achievements such as projects, internships, certificates, etc, will be displayed. Students can show their achievements in the profile where skills will be used as keywords, which is helpful when faculty members search for the skill. The achievements will be displayed in categories. Faculty members can visit the students' profiles.

2. Show Achievements:

To gather all the projects in one place to show the academic works done by and outside skills of a student. So when a faculty will go through a student's profile, this will help them write a strong recommendation letter for the student.

Workflow: A student can share their project, internships, and other skills in the profile. When the student will upload any achievement, the student can request a faculty member for verification of that achievement. It may then be verified by that faculty member.

3. Events:

Events refer to events by clubs, programming contests, concerts, seminars/webinars, etc., where mass participation exists.

Workflow:

The moderators can create events and tag the students that participated, who, in turn, will receive notification about this event. If they accept the tag, it will be added to their profile. Only the faculty members, moderator and the clubs' representatives can create the events. Regular students won't be able to do this.

4. Tag:

Events will be created by admins. The admins may 'tag' students who were part of these events to send them a notification to accept the achievement. Only the people who create events will be able to tag others.

5. Notification:

Whenever a new event will be added, a notification will be sent to the tagged people in the event for them to add that achievement to their profile. Students can request faculty members to approve an achievement, in which case a notification will be sent to the faculty member.

6. Search:

Faculty members will be able to search individual students, events, skills (programming languages, sports, cultural activities, etc), courses, etc. From there, they may tag the students, or just view the profile information. Students will be able to search teachers and events. From there, students may request faculties to add them in an event as a participant.

7. Achievement Verification:

Students can 'tag' a faculty member for verifying their achievement. A 'notification' will reach the faculty. The faculty can look into the achievement and verify it (maybe via a tick mark).

8. Data Control:

The user has full control of their data. They can delete any data of theirs whenever they want, and our servers won't retain the data.

9. Generate CV:

Students can generate their CV from profile. Here bio and all of their achievements and events participation will be shown in categories.

5.4 Tools and Technologies

1. Figma
2. JIRA

3. Fontawesome
4. GitKraken
5. Github

6 JIRA Road Map and Work Distribution

Road Map:

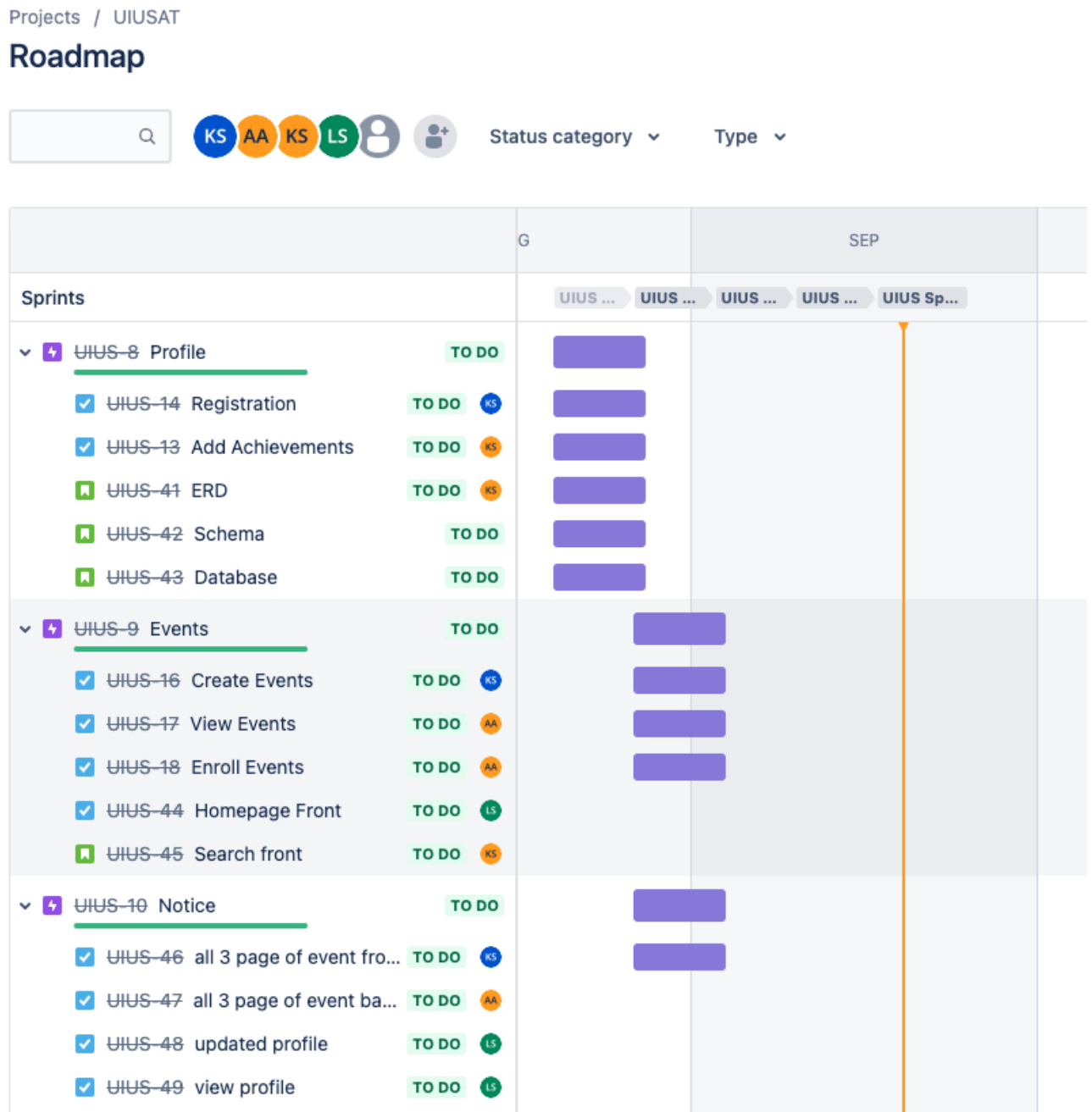


Figure 15: JIRA Road Map part 1

Roadmap

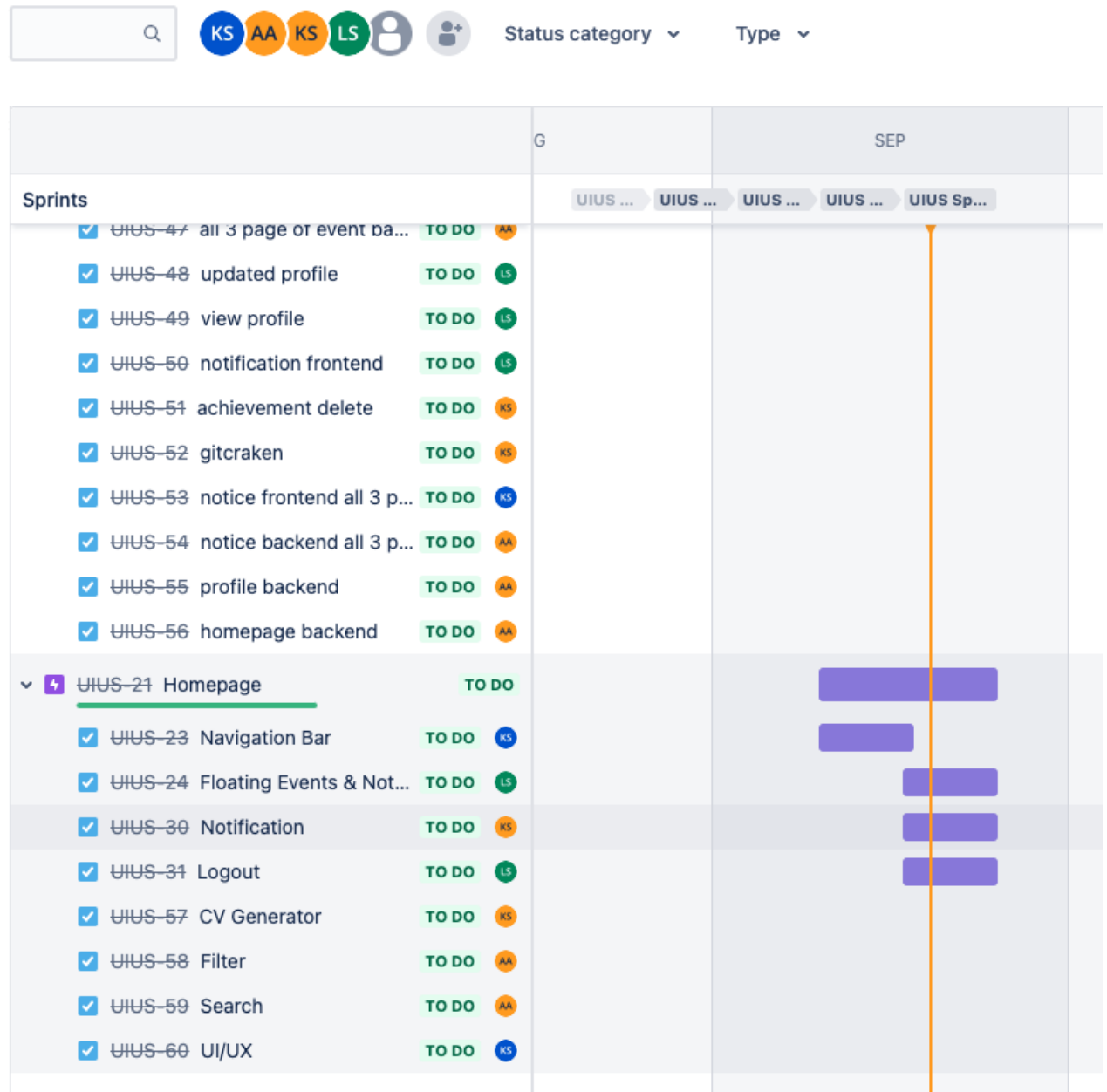


Figure 16: JIRA Road Map part 2

Work Distribution:

Work distribution for each of our teammates for four weeks in given below:

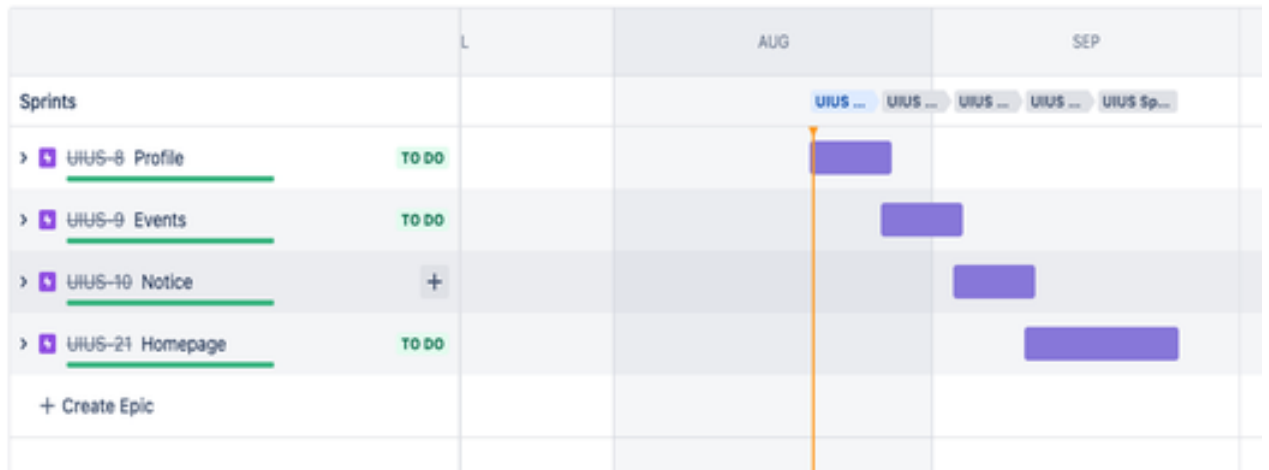


Figure 17: JIRA for 1st week

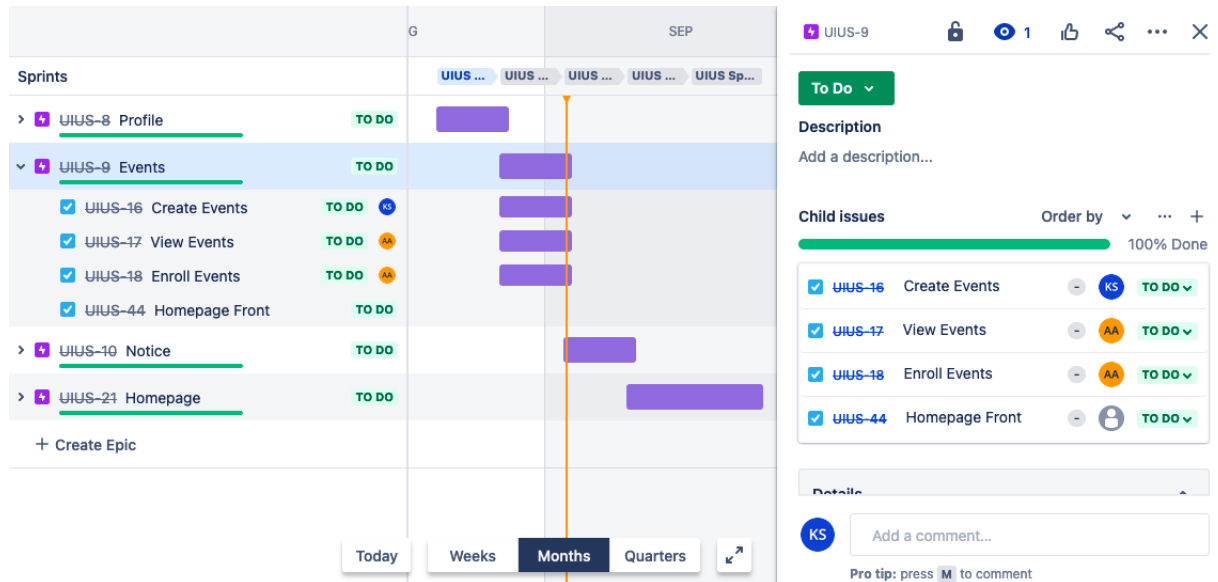


Figure 18: JIRA for 2nd week

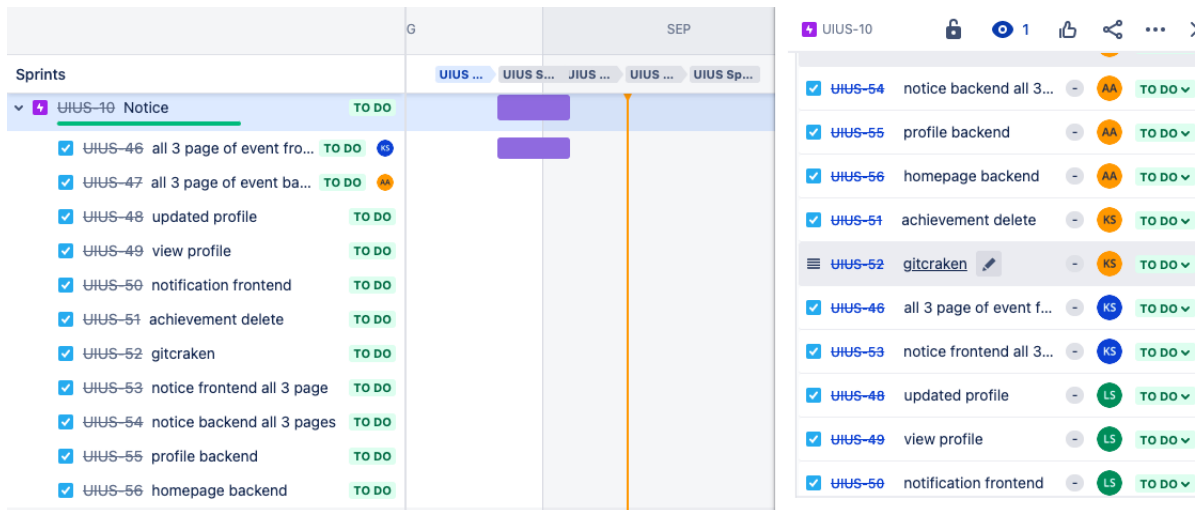


Figure 19: JIRA for 3rd week

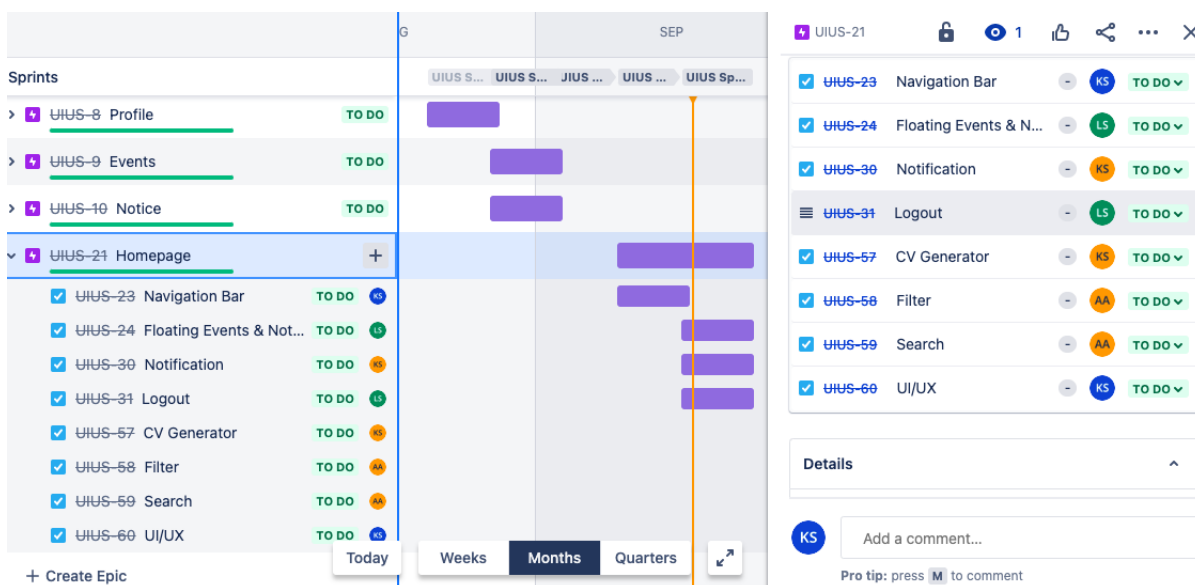


Figure 20: JIRA for 4th week

7 Final Project Update

7.1 Project Completion Status

Our project is fully completed and we implemented all the features mentioned above. This is the github link of our project: <https://github.com/KShawpnil/UIU-Student-Activity-Tracker>

7.2 Requirement Verification

Yes, our requirements are correct and complete, they meet the quality standards defined for them. Project testing is all well and we deployed our project in the project showcase of Summer'22 competition.

7.3 Client/User feedback

All of our users (faculty and students) were very satisfied with our final project. They expressed that they liked our UI design a lot. They also said that the website was really easy to navigate and that they understood the functions well.

7.4 Project Show Feedback

The judges and the other faculties that saw our project were really satisfied with it. They liked our features a lot. We became the 1st Runner Up in the Project Show.

8 Conclusions and Future Work

To conclude, this document is a comprehensive and complete set of guidelines for how the system should be made. Sticking to it as closely as possible will help us in reaching the target audience. This document takes you through all the steps taken to design the system and gives a clear picture of what the requirements are of the company.

8.1 Future Work:

In the future we plan to work under Dr. Swakkharto Shatabda to make an official website for this project. We also plan to improve some of our features to make it easier for faculties to use it. We will also try to link UCAM with our website, so that we can get all the CGPA and other information of the students' in website.